

GUI Explanation for Professor - Bearing Fault Diagnosis Project

1. Purpose of the GUI

This GUI demonstrates bearing fault diagnosis in a practical way. It takes vibration signal data as input, extracts important features, and uses the trained machine learning model to predict the bearing condition.

2. Input Settings

Sampling Rate (Hz): Number of amplitude samples recorded per second. It is needed for time scaling and

Upload Vibration CSV: Uploads a CSV file containing vibration amplitude values.

Use Built-In Demo Signal: Tests the system using a prepared example signal.

Expected Classes: Normal, Inner Race Fault, Outer Race Fault, Ball Fault.

3. Workflow Section

A vibration signal is uploaded. Then the system computes time-domain and FFT-based features and sends them to the trained machine learning model for classification.

4. Predicted Fault

This shows the final bearing condition predicted by the trained model.

5. Confidence

This is the probability score of the predicted class. Higher value means the model is more confident.

6. Signal Length

This shows how many amplitude samples are present in the uploaded signal window.

7. Time-Domain Waveform

This graph shows amplitude versus time and represents the raw vibration behavior of the bearing signal.

8. Frequency Spectrum (FFT)

This graph shows magnitude versus frequency after FFT and helps identify dominant frequency components.

9. Confidence by Fault Type

This table shows the probability of the signal belonging to each fault category.

10. Extracted Features

This section shows numerical features extracted from the vibration signal, such as mean, standard deviation, RMS, kurtosis, spectral mean, and frequency variation factor. These features are used as input to the classifier.

11. Confusion Matrix

This shows model performance on test data. Diagonal values are correct predictions and off-diagonal values are misclassifications.

12. Feature Importance

This shows which extracted features are most important for the Random Forest model.

13. Backend Flow

After uploading the CSV, the system reads the amplitude values, extracts features, passes them to the trained model, predicts the most likely fault type, and displays the result in the GUI.

14. Short Overall Explanation

This GUI is the practical front-end of the bearing fault diagnosis system. It takes vibration data as input, extracts meaningful features, classifies the bearing condition using a trained ML model, and displays the result along with signal plots and model interpretation outputs.